

Build the sampled spectrum

Background

Use angular frequency w in rad/s. Let $X(w) = \max(1 - |w|/W, 0)$, where $W > 0$. Sampling with period T creates $X_p(w) = (1/T) \sum X(w - k \omega_s)$, where $\omega_s = 2\pi/T$. In this task the displayed sum contains only $k = -K, \dots, K$. Overlapping copies add.

Your task

Return the original triangular spectrum and the finite sum of shifted, scaled copies. Display the supplied two-case plot for separated and overlapping copies. Include all requested copies, even when a copy extends outside the displayed frequency range.

Function contract and assumptions

`base_spectrum(w, W)` returns $X(w)$. `sampled_spectrum(w, W, T, K)` returns $X_p(w)$ using exactly $2K+1$ copies. Outputs match the shape of w . Assume $W, T > 0$ and integer $K \geq 0$. Do not calculate a Fourier transform.

Example / correctness check

With $W=2$, $T=\pi$ and $K=1$, $\omega_s=2$. At $w=[-2, 0, 2]$, the finite sampled spectrum equals $[1/\pi, 1/\pi, 1/\pi]$. The original spectrum at those points is $[0, 1, 0]$.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 16-22.